

1. Falling onto each other: Consider two particles which are moving under the influence of their mutual gravitational force. Let the particles follow circular orbits about one another with a time period T . Show that, if the particles are suddenly stopped in their orbits and allowed to gravitate towards each other, they will collide after a time $T/(4\sqrt{2})$.

$$T^2 = \frac{4\pi^2 \mu r_0^3}{\alpha} \quad \text{and when stopped, } E = \frac{1}{2} \mu \dot{r}^2 + \cancel{\frac{l^2}{2mr^2}} + V(r)$$

$$\sqrt{E + \frac{\alpha}{r}} = \sqrt{\frac{\mu}{2}} \frac{dr}{dt} \quad \text{with } E = -\frac{\alpha}{r_0}$$

$$\Rightarrow \sqrt{\frac{\alpha}{r} - \frac{\alpha}{r_0}} = \sqrt{\frac{\mu}{2}} \frac{dr}{dt}$$

$$\int_0^T dt = \sqrt{\frac{\mu}{2\alpha}} \int_{r_0}^0 \frac{\sqrt{r_0 r}}{\sqrt{r_0 - r}} dr$$

$$r = r_0 \sin^2 \theta \Rightarrow dr = 2r_0 \sin \theta \cos \theta d\theta$$

$$T = \sqrt{\frac{\mu}{2\alpha}} \int_{\frac{\pi}{2}}^0 2r_0^{3/2} \sin^2 \theta d\theta = r_0^{3/2} \sqrt{\frac{\mu}{2\alpha}} \frac{\pi}{2} \Rightarrow T^2 = \frac{\pi^2}{8\alpha} \mu r_0^3$$

$$\frac{T^2}{T^2} = \frac{1}{32} \Rightarrow$$

$$\boxed{T = \frac{T}{4\sqrt{2}}}$$

2. Precessing orbits: Show that the motion of a particle in the central potential

$$U(r) = -\frac{\alpha}{r} + \frac{h}{r^2},$$

with $\alpha > 0$, is the same as that of motion under the Kepler potential alone when expressed in terms of a coordinate system rotating or precessing around the center of the force.

$$E = \frac{1}{2} m \dot{r}^2 - \frac{\alpha}{r} + \frac{(h + l^2/2m)}{r^2}$$

$$\text{let } \frac{h + l^2}{2m} = \frac{l'^2}{2m}$$

$$E = \frac{1}{2} m \dot{r}^2 - \frac{\alpha}{r} + \frac{l'^2}{2mr^2}$$

$$E + \frac{\alpha}{r} - \frac{l'^2}{2mr^2} = \frac{m}{2} \left(\frac{dr}{dt} \right)^2 = \left(\frac{dr}{d\phi} \right)^2 \frac{l^2}{2m r^4}$$

$$E + \frac{\alpha}{r} - \frac{l'^2}{2mr^2} = \frac{l^2}{2m} \left(\frac{1}{r^2} \frac{dr}{d\phi} \right)^2$$

$$u = 1/r$$

$$E + \alpha u - \frac{l'^2}{2m} u^2 = \frac{l^2}{2m} \left(\frac{du}{d\phi} \right)^2$$

$$\begin{aligned} & \sqrt{\frac{2mE}{l^2} + \frac{2m\alpha}{l^2} u - \frac{l'^2}{l^2} u^2} = \frac{du}{d\phi} \\ & - \left(\frac{l'^2}{l^2} u^2 - \frac{2m\alpha}{l^2} u - \frac{2mE}{l^2} \right) \\ & = - \left(\left(\frac{l'u}{l} - \frac{m\alpha}{l'l} \right)^2 - \frac{m^2 \alpha^2}{l'^2 l^2} - \frac{2mE}{l^2} \right) \\ & = \frac{2mE}{l^2} + \frac{m^2 \alpha^2}{l'^2 l^2} - \left(\frac{l'u}{l} - \frac{m\alpha}{l'l} \right)^2 \end{aligned}$$

$$d\phi = \frac{du}{\sqrt{\frac{2mE}{l^2} + \frac{m^2 \alpha^2}{l'^2 l^2} - \left(\frac{l'u}{l} - \frac{m\alpha}{l'l} \right)^2}}$$

A^2

$$\frac{l'u}{l} - \frac{m\alpha}{l'l} = A \sin p$$

$$\Rightarrow \frac{l'}{l} du = A \cos p dp$$

$$d\phi = \frac{l'}{l} dp \Rightarrow \sin \left(\frac{l}{l'} (\phi - \phi_0) \right) = \frac{l'u}{Al} - \frac{m\alpha}{Al'l}$$

$$u = \frac{m\alpha}{l'^2} + l A \sin \left(\frac{l'}{l} (\phi - \phi_0) \right)$$

$$\frac{1}{r} = \frac{m\alpha}{l'^2} + \sqrt{2mE + \frac{m^2\alpha^2}{l'^2}} \sin\left(\frac{l'}{l}(\phi - \phi_0)\right)$$

$$\left\{ \frac{l'^2}{2m} = \frac{l^2}{2m} + h \right\}$$

Now $l'^2 = l^2 + 2mh$

$$\frac{l'^2}{l^2} = 1 + \frac{2mh}{l^2} \Rightarrow \frac{l'}{l} \approx 1 + \frac{mh}{l^2}$$

Say $r=r_0$ at $\phi=\phi_0$, then $r=r_0$ again when

$$\frac{l'}{l}(\phi - \phi_0) = 2\pi$$

$$\Rightarrow \phi = \phi_0 + 2\pi \frac{l}{l'} = \phi_0 + 2\pi + \frac{2\pi mh}{l^2}$$

instead of
 $\phi_0 + 2\pi$
↑

So the orbit slightly precesses by

$$\frac{2\pi mh}{l^2}$$

3. Precession of the perihelion of Mercury: When a small correction $\delta U(r)$ is added to the potential energy $U = -\alpha/r$, with $\alpha > 0$, the paths of finite motion are no longer closed and in each revolution the perihelion is displaced through a small angle, say, $\delta\phi$.

- (a) Find $\delta\phi$ when: (i) $\delta U = \beta/r^2$ and (ii) $\delta U = \gamma/r^3$. Compare the result for the first case with the result from the previous exercise.
- (b) For the case of the planet Mercury, the value of the semi-major axis of its orbit is $a = 5.79 \times 10^{10}$ m and its eccentricity is $e = 0.206$. Also, the period of the Mercury's orbit around the Sun is 88 days. Further, the value of the gravitational constant is $G = 6.673 \times 10^{-11} \text{ kg}^{-1} \text{ m}^3 \text{ s}^{-2}$, the speed of light is $c = 2.998 \times 10^8 \text{ m/s}$, and the mass of the Sun is $M_\odot = 1.989 \times 10^{30} \text{ kg}$. Estimate the extent of precession of Mercury's orbit if $\delta U = -\alpha \ell^2 / (m^2 c^2 r^3)$, where $\alpha = G M m$, ℓ is the angular momentum, and c is the velocity of light.

Note: The δU mentioned above arises due to general relativistic effects. The measured precession of the perihelion of the planet Mercury turns out to be $5599''.7 \pm 0''.4$ per century, but a large part of it is caused due to the influences of the other planets. When the other contributions have been subtracted, the precession of the perihelion of the planet Mercury due to the purely general relativistic effects amounts to 43.1 ± 0.5 seconds of arc per century.

$$E = \frac{L^2}{2mr^4} \left(\frac{dr}{d\phi} \right)^2 + \frac{L^2}{2mr^2} + V$$

$$2m(E - V) - \frac{L^2}{r^2} = \frac{L^2}{r^4} \left(\frac{dr}{d\phi} \right)^2$$

$$\sqrt{2m(E - V) - \frac{L^2}{r^2}} = \frac{L}{r^2} \frac{dr}{d\phi}$$

$$\Rightarrow \int d\phi = \int \frac{L/r^2}{\sqrt{2m(E - V) - \frac{L^2}{r^2}}} dr$$

$$\Delta\phi = -2 \frac{\partial}{\partial L} \int \sqrt{2m(E - V) - \frac{L^2}{r^2}} dr$$

$$V \mapsto V + \delta V$$

$$\Delta\phi' = -2 \frac{\partial}{\partial L} \int \sqrt{2m(E - V) - \frac{L^2}{r^2} - 2m\delta V} dr$$

$$\Delta\phi' = -2 \frac{\partial}{\partial L} \int \sqrt{2m(E - V) - \frac{L^2}{r^2}} \left(1 - \frac{m\delta V}{2m(E - V) - \frac{L^2}{r^2}} \right) dr$$

$$\Delta\phi' = -2\frac{\partial}{\partial L} \int \sqrt{2m(E-V) - \frac{L^2}{r^2}} + 2\frac{\partial}{\partial L} \int \frac{m \delta V}{\sqrt{2m(E-V) - \frac{L^2}{r^2}}} dr$$

$$= \Delta\phi + \delta\phi \quad \text{and} \quad 2m(E-V) - \frac{L^2}{r^2} = \frac{L^2}{r^2} \left(\frac{dr}{d\phi} \right)^2$$

$$\delta\phi = 2\frac{\partial}{\partial L} \int_0^\pi \frac{r^2 m}{L} \delta V d\phi$$

$$i) \delta V = \frac{\beta}{r^2} \Rightarrow \delta\phi = 2\frac{\partial}{\partial L} \int_0^\pi \frac{m\beta}{L} d\phi = \boxed{\frac{2\pi\beta m}{L^2}}$$

$$ii) \delta V = \frac{\gamma}{r^3} \Rightarrow \delta\phi = 2\frac{\partial}{\partial L} \int_0^\pi \frac{\gamma m}{L^3} (1 + e \cos\phi) d\phi$$

$$\delta\phi = 2\frac{\partial}{\partial L} \int_0^\pi \frac{\gamma m^2 \alpha}{L^3} (1 + e \cos\phi) d\phi$$

$$\boxed{\delta\phi = \frac{6\gamma m^2 \alpha \pi}{L^4}}$$

$m \rightarrow$ red. mass

$$b) \text{ For mercury, } \gamma = -\frac{\alpha L^2}{m^2 c^2} \quad m \rightarrow \text{mercury mass}$$

$$\delta\phi = \frac{6\alpha L^2}{m^2 c^2} \frac{\mu^2 \alpha \pi}{L^4}$$

$$= \frac{6\alpha^2 \mu^2 \pi}{m^2 c^2 L^2}$$

$$= \frac{6\mu\pi\alpha}{m^2 c^2 a(1-e^2)}$$

$$\delta\phi = \frac{6GM^2\pi}{(M+m)c^2 a(1-e^2)} \quad (\text{take } m \ll M)$$

$$a = 5.79 \times 10^{10} \text{ m}$$

$$e = 0.206$$

$$G = 6.673 \times 10^{-11}$$

$$c = 2.998 \times 10^8$$

$$M = 1.989 \times 10^{30}$$

$$T = 88 \text{ days}$$

$$e^2 = 1 + \frac{2EL^2}{\mu \alpha^2}$$

$$\frac{\mu \alpha^2 (e^2 - 1)}{2E} = L^2$$

$$L^2 = \mu \alpha a(1-e^2)$$

radians

$$\Rightarrow \delta\phi = 5.02 \times 10^{-7} \text{ } ^{\wedge} \text{ for one Mercury year} = \text{for } \frac{88}{365} \text{ Earth years.}$$

$$\Rightarrow \boxed{\delta\phi \approx 42.947} \text{ arc seconds per Earth century}$$

4. Motion in the Yukawa potential: A particle moves in the Yukawa potential described by

$$V(r) = -\frac{k}{r} \exp -(r/a),$$

where k and a are positive quantities.

- (a) Write down the equations of motion and reduce them to the equivalent one-dimensional problem.
- (b) Use the effective potential to discuss the qualitative nature of the orbits for different values of the energy and the angular momentum.
- (c) Also, show that if the orbit is nearly circular, the apsides will advance approximately by $\pi \rho^2/a^2$ per revolution, where ρ is the radius of the circular orbit.

Note: The radial distances to the turning points are known as apsidal distances.

5. Eliminating the centre of mass: A system consists of one particle of mass M and n particles with equal masses m . Eliminate the motion of the centre of mass and thereby reduce the problem to one involving n particles.

$$M: \bar{R} \quad m: \bar{R}_a \quad \bar{r}_a = \bar{R}_a - \bar{R} \quad \dot{\bar{R}}_a = \dot{\bar{r}}_a + \dot{\bar{R}}$$

let origin be at CoM.

$$M\bar{R} + m \sum \bar{R}_a = 0 \Rightarrow (M+nm)\bar{R} + m \sum \bar{r}_a = 0$$

$$\Rightarrow \bar{R} = \frac{-m}{M+nm} \sum \bar{r}_a$$

$$\mathcal{L} = \frac{1}{2} M \dot{\bar{R}}^2 + \frac{1}{2} m \sum \dot{\bar{R}}_a^2 - U$$

$$\dot{\bar{R}}_a^2 = \dot{\bar{r}}_a^2 + \dot{\bar{R}}^2 + 2\dot{\bar{r}}_a \cdot \dot{\bar{R}}$$

$$\bar{R}_a = \bar{r}_a + \bar{R}$$

$$\dot{\bar{R}}_a = \dot{\bar{r}}_a + \dot{\bar{R}}$$

$$\dot{\bar{R}}_a^2 = \dot{\bar{r}}_a^2 + \dot{\bar{R}}^2 + 2\dot{\bar{r}}_a \cdot \dot{\bar{R}}$$

$$\bar{R} = \frac{-m}{M+nm} \sum \bar{r}_a$$

$$\dot{\bar{R}} = \frac{-m}{M+nm} \sum \dot{\bar{r}}_a$$

$$\dot{\bar{R}}_a^2 = \dot{\bar{r}}_a^2 + \frac{m^2}{(M+nm)^2} (\sum \dot{\bar{r}}_a)^2 - \frac{2m}{M+nm} \dot{\bar{r}}_a \cdot \sum \dot{\bar{r}}_a$$

$$\mathcal{L} = \frac{1}{2} \frac{M m^2}{(M+nm)^2} (\sum \dot{\bar{r}}_a)^2 + \frac{1}{2} m \sum \dot{\bar{r}}_a^2 + \frac{n m^3}{2(M+nm)^2} (\sum \dot{\bar{r}}_a)^2$$

$$- \frac{m^2}{(M+nm)} \sum \dot{\bar{r}}_a \cdot \sum \dot{\bar{r}}_a - U$$

$$\mathcal{L} = \frac{m^2}{2(M+nm)} (\sum \dot{\bar{r}}_a)^2 + \frac{1}{2} m \sum \dot{\bar{r}}_a^2 - \frac{m^2}{(M+nm)} \sum \dot{\bar{r}}_a \cdot \sum \dot{\bar{r}}_a - U$$

Notice that $\sum \dot{\bar{r}}_a \cdot (\sum \dot{\bar{r}}_a) = (\sum \dot{\bar{r}}_a) \cdot (\sum \dot{\bar{r}}_a) = (\sum \dot{\bar{r}}_a)^2$

$$\Rightarrow \boxed{\mathcal{L} = \frac{m}{2} \sum \dot{\bar{r}}_a^2 - \frac{m^2}{2(M+nm)} (\sum \dot{\bar{r}}_a)^2 - U}$$